Golden tilefish (*Lopholatilus chamaeleonticeps*)

Summary Scores

- **Sensitivity:** Moderate
- **Exposure:** Moderate
- **Vulnerability:** Moderate

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	2.96	2.53	
Complexity in Reproductive Strategy	1.84	2.04	
Dispersal of Early Life Stages	2.28	2.05	
Egg and Larval Survival and Settlement Requirements	2.24	2.22	
Historical Variability in Temperature	2.72	2.71	
Juvenile Habitat Requirements	1.24	2.9	
Other Stressors	1.52	2.46	
Population Growth Rate	2.8	2.6	
Prey Specificity	1.48	2.82	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.52	2.6	
Spawning Cycle	2.28	2.8	
Stock Size/Status	3	2.8	
Total Sensitivity | Certainty	**Moderate (2) | 0.62**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	1.08	0.21	
Bottom Oxygen	1.64	0.54	
Bottom Salinity	3.51	0.54	
Bottom Aragonite Solubility	2.47	1.20	
Net primary production	1	0.02	
Downward Flux of Particulate Organic Carbon	1.02	0.06	
Total Exposure	**Moderate (1.87)**	**0.94**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Moderate (3.75)**	**1.88**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Golden tilefish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	High (Burton et al. 2023)	Moderate	Decrease
Exposure	Very High (Burton et al. 2023)	Moderate	Decrease
Vulnerability	Very High (Burton et al. 2023)	Moderate	Decrease
Directional Effect	Neutral (Burton et al. 2023)	Neutral	None

Stocks

Golden tilefish is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

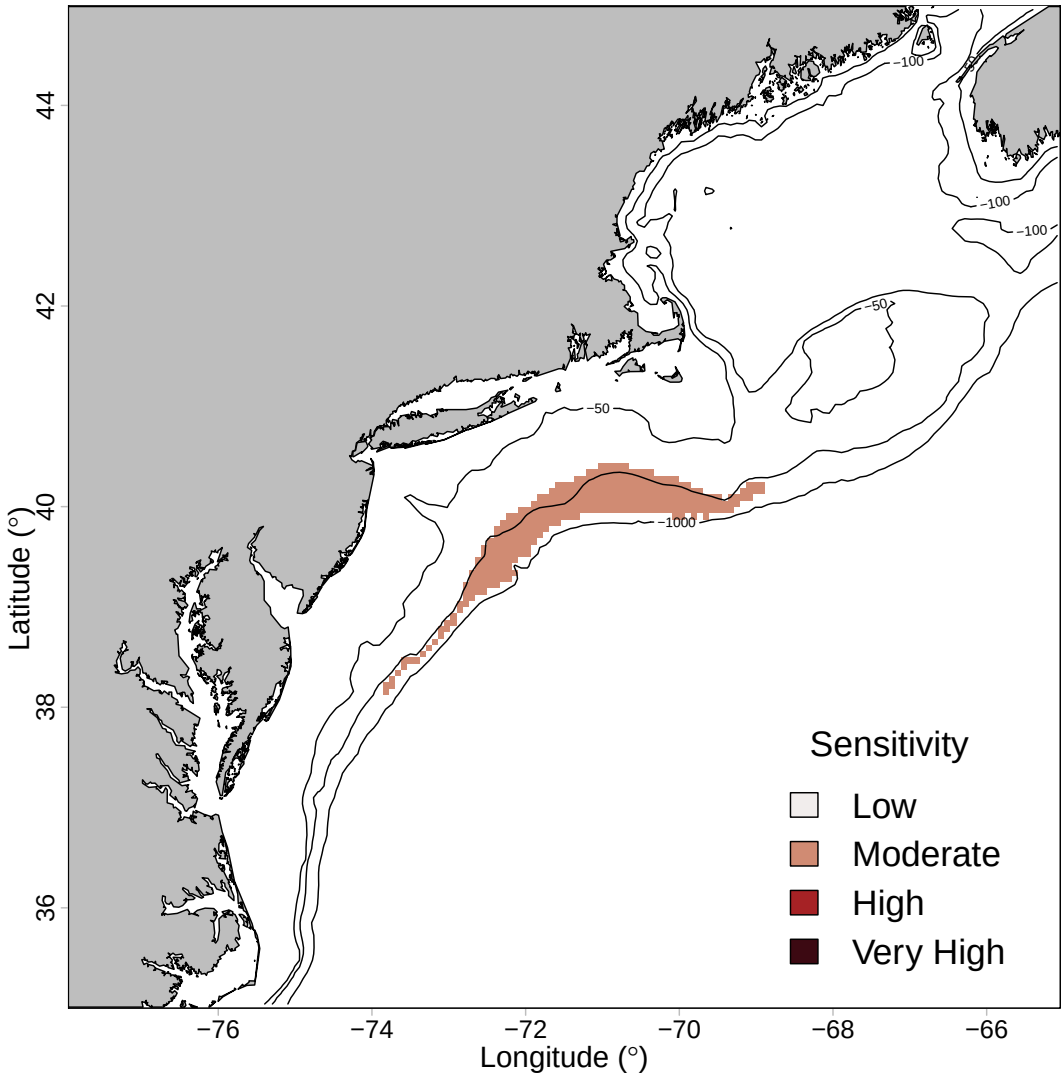


Figure 2: Golden tilefish Sensitivity on the Northeast Shelf

Golden Tilefish received a Moderate sensitivity score with 62% certainty (Figure 1). Four attributes had scores greater than 2.5: Stock Size (3), Adult Mobility (3), Population Growth Rate (2.8), and Historical Variability in Temperature (2.7) (Figure 1).

Golden Tilefish primarily occupy burrows on the seafloor. Die offs have been observed after meanders in the Gulf Stream, suggesting that they are very sensitive to temperature shifts (Able et al. 1982). Golden tilefish mature at 2-4 years and maximum ages of 45 years have been documented (Steimle 1999) (Bigelow and Schroeder 1953), suggesting they are a long-lived species that may be susceptible to overfishing. Overfishing is currently occurring in the Mid-Atlantic stock (NEFSC 2024).

Golden tilefish were not assessed along the Northeast Shelf in the 2016 Assessment, but were assessed in the recent Southeast US Climate Vulnerability Assessment. They received a High sensitivity score in

that assessment. This was driven by two attributes: Habitat Specialization and Population Growth Rate (Burton et al. 2023), similar to our results.

Exposure:

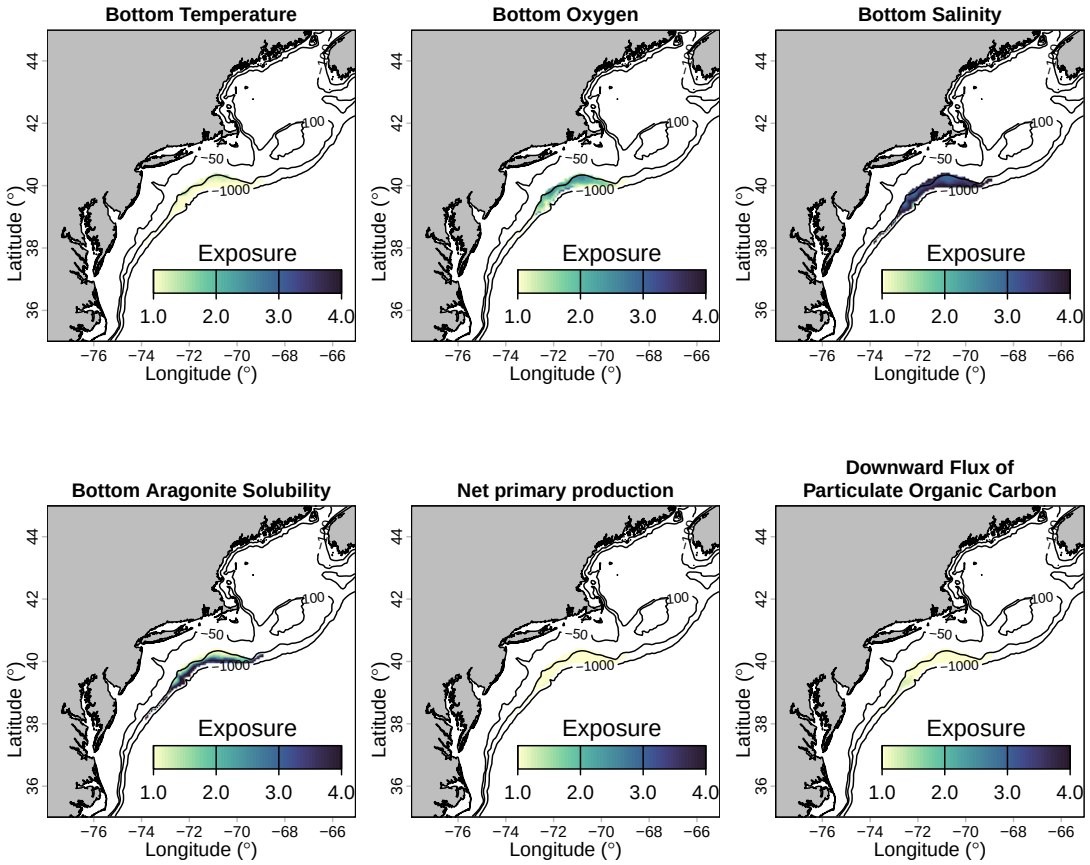


Figure 3: Maps of each Exposure Factor for Golden tilefish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Golden Tilefish was Moderate (1.87 +/- 0.94) (Figure 1). Exposure was greatest at the shelf break and decreased on the shelf (Figure 4). This was driven by high exposure to changes in bottom oxygen, salinity, and aragonite solubility (Figure 3).

Golden tilefish received an Exposure score of Very High in the recent Climate Vulnerability Assessment for the Southeast Shelf (Burton et al. 2023). This score was driven by high exposure to sea surface temperature, salinity, ocean acidification, and currents (Burton et al. 2023).

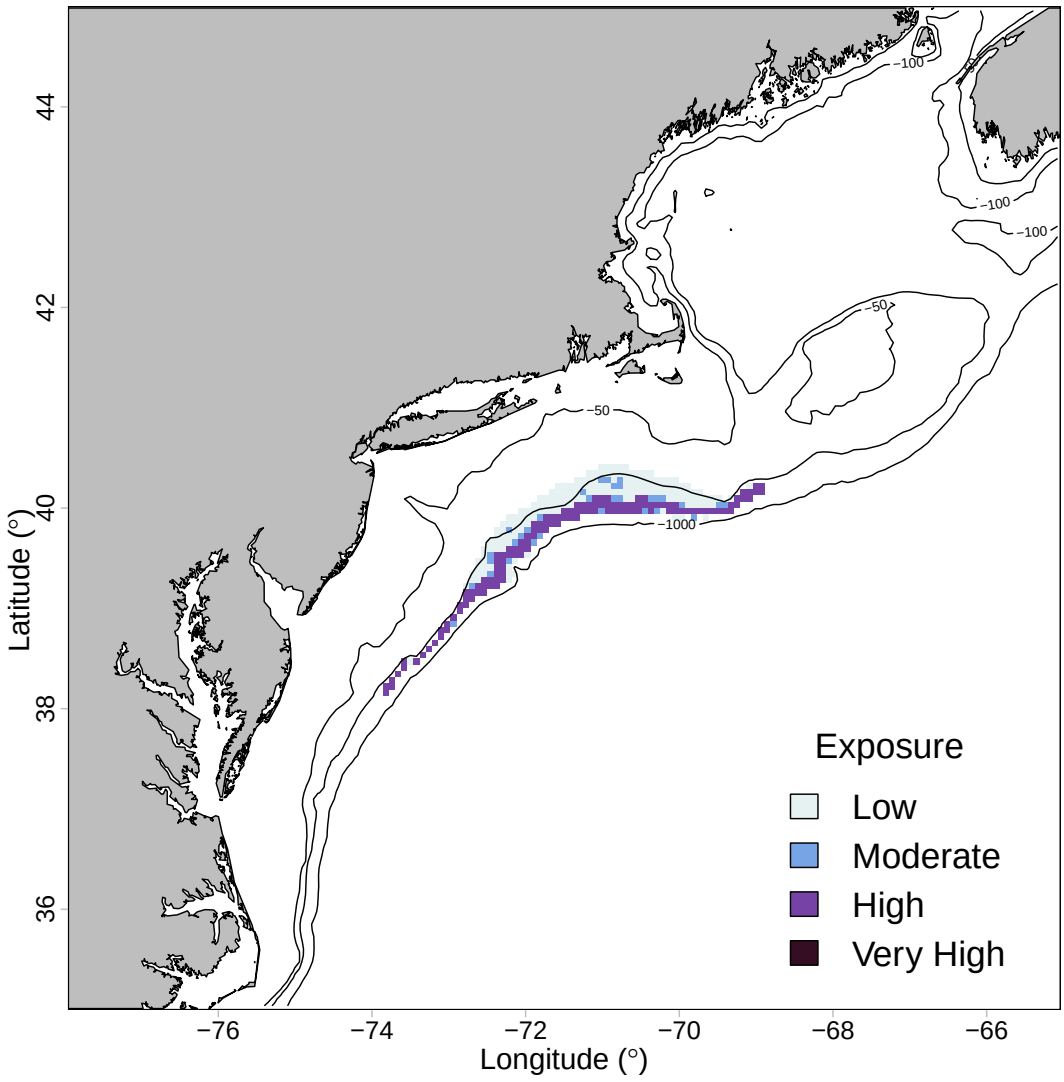


Figure 4: Total Exposure on the Northeast Shelf for Golden tilefish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

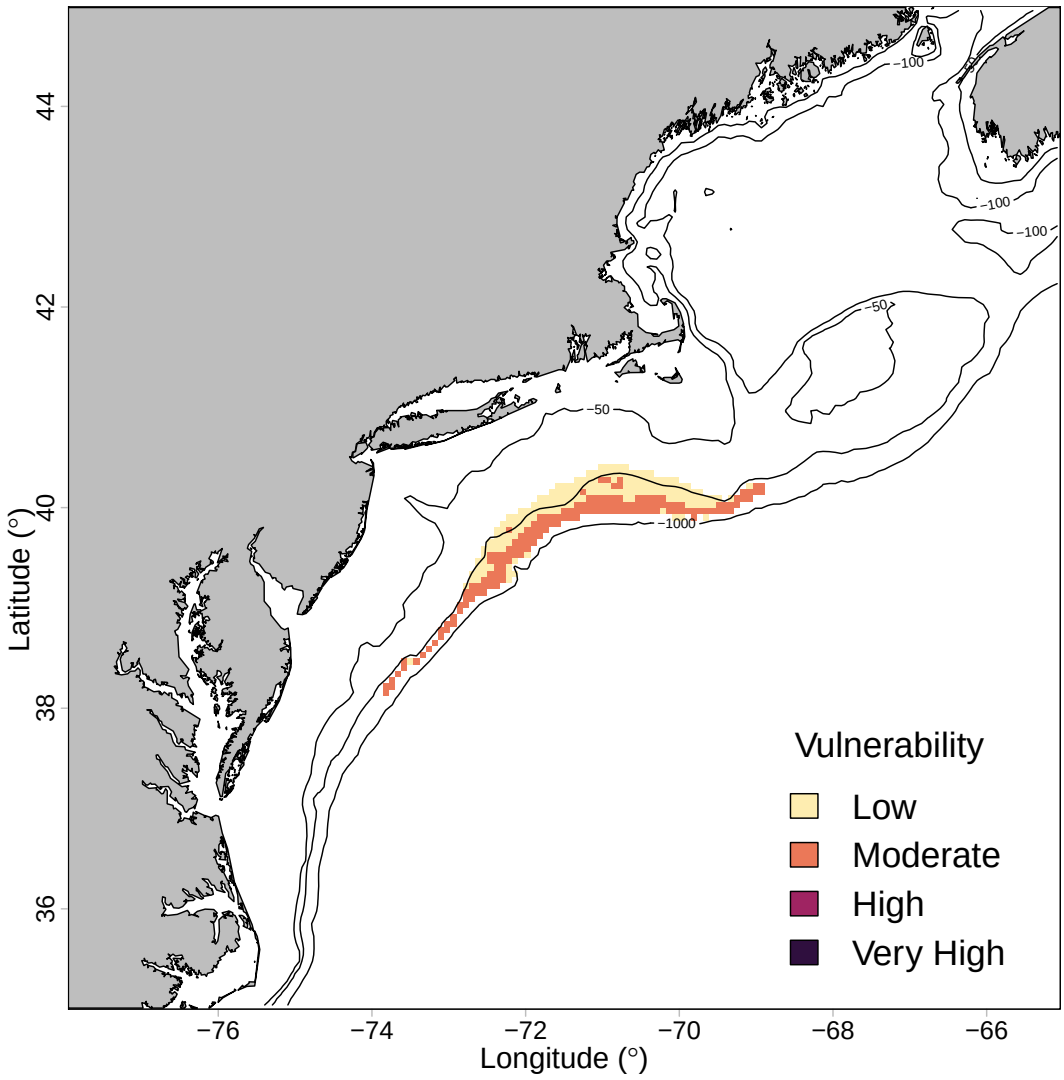


Figure 5: Vulnerability on the Northeast Shelf for Golden tilefish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean overall Vulnerability for Golden Tilefish along the Northeast Shelf was Moderate (3.75 +/- 1.88) (Figure 1). Scores were Moderate along the shelf break and decreased to Low along the continental shelf (Figure 5). When only important drivers of species distribution were used in the exposure calculation, Vulnerability decreased to Low (2.12 +/- 0.59) (Figure 1).

In the recent assessment for the Southeast Shelf, Golden Tilefish received a vulnerability score of Very High (Burton et al. 2023).

Data Quality Metrics

Life History Data Quality:

93% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

Mean model confidence for golden tilefish was 2.3 (+/- 0.57). Similar to blueline tilefish, experts stated that the models accurately represented known distributions of golden tilefish, but low sample sizes due to catchability drove down confidence. The model AUC was 0.95, indicating excellent performance.

Additional Metrics

Distribution Shifts:

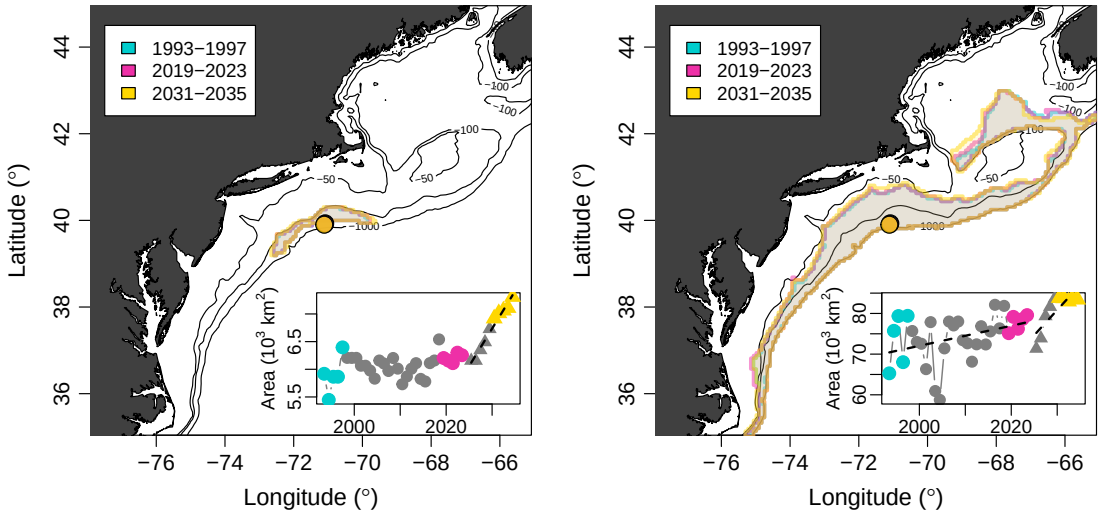


Figure 6: Distribution shifts on the Northeast Shelf for Golden tilefish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Golden tilefish range expanded significantly between 1993 and 2023 due to increased habitat availability in the Gulf of Maine and along the continental shelf (Figure 6). The weighted centroid shifted 2 km south-southwest.

Under near future climate conditions, these trends are predicted to continue. Both core habitat and total range of golden tilefish is predicted to expand significantly. Core habitat is predicted to expand laterally along the continental shelf break and the species' range is predicted to continue to expand onto

the continental shelf and into the Gulf of Maine (Figure 6). The weighted centroid is predicted to shift a further 2 km southwest.

In the previous assessment for the southeast shelf, golden tilefish received a Moderate Distributional Vulnerability Rank due to their sensitivity to temperature, adult mobility, and larval dispersal (Burton et al. 2023).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	5,968	73,613	2.62	5.53
2019-2023	2.0 km SSW	6,205 (+237)	77,943 (+4,329)	2.81 (+0.19)	5.69 (+0.16)
2031-2035	2.3 km SW	7,061 (+856)	83,590 (+5,647)	3.24 (+0.44)	5.93 (+0.24)

Directional Effect:

Experts predicted that climate change would have a neutral impact on Golden tilefish. Certainty was 92%. Golden Tilefish occupy the 'warm belt' where shelf and slope waters meet (Able et al. 1982). Warming temperatures could increase the amount of preferred habitat. In the Southeast Climate Vulnerability Assessment, experts also predicted that climate change will have a neutral impact on Golden Tilefish, and hypothesized that warming temperatures could increase the amount of available habitat in the northeast (Burton et al. 2023).

Climate Effects on Abundance and Distribution

Golden tilefish prefer a very narrow temperature range (8-17°C); increased ocean acidification may have negative impacts on survival, growth, and abundance (Salois et al. 2024). Nesslage et al. (2021) identified a correlation between negative Atlantic Multidecadal Oscillation (AMO) and positive North Atlantic Oscillation (NAO) and golden tilefish recruitment class pulses. Catch per unit effort is influenced by temperature and oceanographic currents (Gulf Stream and Labrador Current) (Nesslage et al. 2021). A large golden tilefish die-off and resulting fishery collapse in the spring of 1882 has been correlated with meanders in the Gulf Stream, suggesting that they are very sensitive to temperature changes (Fisher et al. 2014). Changes to weather patterns and a rise in stormy conditions along with very cold bottom temperatures can drive golden tilefish off the shelf and cause a decrease in fishing (Salois et al. 2024).

Life History Synopsis

Golden tilefish inhabit continental shelf and slope waters along the entire U.S. East and Gulf Coasts, although they are most abundant from Nantucket to Cape May, New Jersey (Steimle 1999). Two genetically-different tilefish stocks have been identified in United States waters: the Mid-Atlantic Bight and south of Cape Hatteras into the Gulf of America (Steimle 1999). Individuals of many species off Cape Hatteras were smaller and less active than their conspecifics off Virginia. Tilefish are classified as serial or fractional spawners from March to November with a peak in activity in June (Idelberger 1988). However,

with serial or fractional spawners there is some doubt whether all of the eggs in the ovaries are released during a single seasonal spawning cycle. Residual eggs could be resorbed during the winter (Grimes et al. 1988). Artificially-fertilized eggs hatched in 40 hours at 22-24.6°C; it is likely that egg duration increases in colder temperatures (Fahay and Berrien 1981). The duration of the larval stage is unknown. Movements of water masses along the shelf break may play a role in recruitment success, including larval survival and retention near spawning grounds. Strong recruitment pulses have been linked to shifts in the Gulf Stream Index, AMO, and NAO (Salois et al. 2024). Little is known about larval diets, but they likely prey on zooplankton, particularly *Calanus* (Steimle 1999). Common predators of larval and juvenile tilefish are monkfish, spiny dogfish, and bottom-dwelling sharks (Steimle 1999). Post-larval tilefish are primarily sedentary and non-migratory. They settle in waters approximately 8-17°C, known as the "warm belt" (where shelf and slope water meet) on the outer continental shelf and upper slope of the Northwest Atlantic (Able et al. 1982). Juveniles and adults often occupy simple vertical shaft burrows in semi-lithified clay of the shelf break, submarine canyon walls, and flanks around 100-200m depth (Able et al. 1982). Lobster pots, red crab traps, ship wrecks and other artificial structures have also been used by tilefish (Steimle 1999). Diet shifts with size, with crustaceans, mollusks, and echinoderms dominating young/small tilefish diets. Adult tilefish are opportunistic benthic foragers, feeding on shrimp, crabs, snails, worms, anemones, sea cucumbers, smaller fish, squid, and bivalves. They are known to scavenge on fresh material and reject rancid bait. Adults will also consume anthropogenic material (Steimle 1999). Age at maturity is typically 2-4 years (Steimle 1999), with a maximum age up to 45 years (Bigelow and Schroeder 1953). The Northwest Atlantic stock is managed by the Mid-Atlantic Fishery Management Council under the Tilefish Fishery Management Plan. The most recent benchmark stock assessment occurred in 2024. The stock is not overfished, but overfishing is occurring and the Mid-Atlantic stock is reduced to end overfishing (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

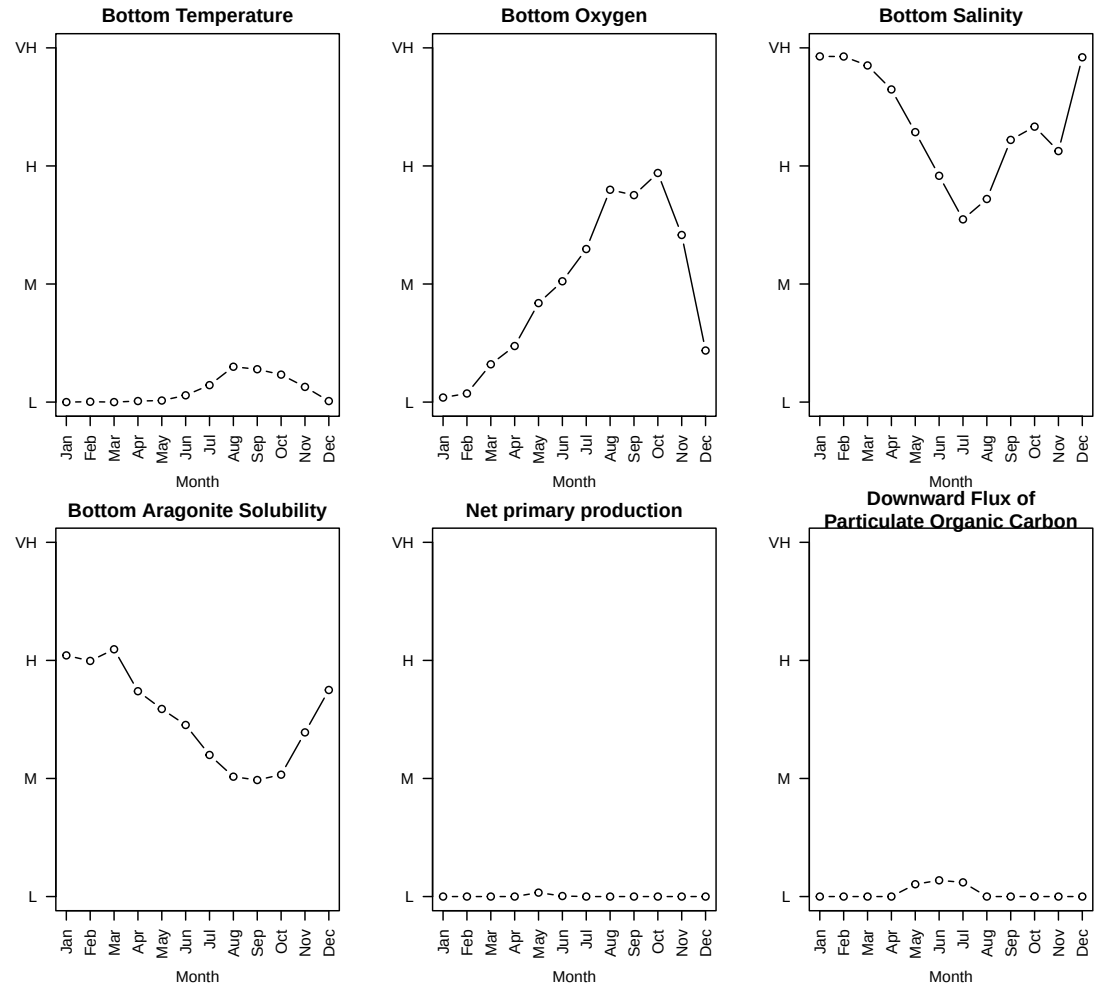


Figure 7: Monthly Timeseries of each Exposure Factor for Golden tilefish. Exposure was calculated between 2014-2023 and 2025-2035

Exposure ranged from High to Low across months. Exposure was generally higher in the winter months (January - March) than the rest of the year (Figure 8). This was due to high bottom salinity and aragonite solubility exposure during those months (Figure 7).

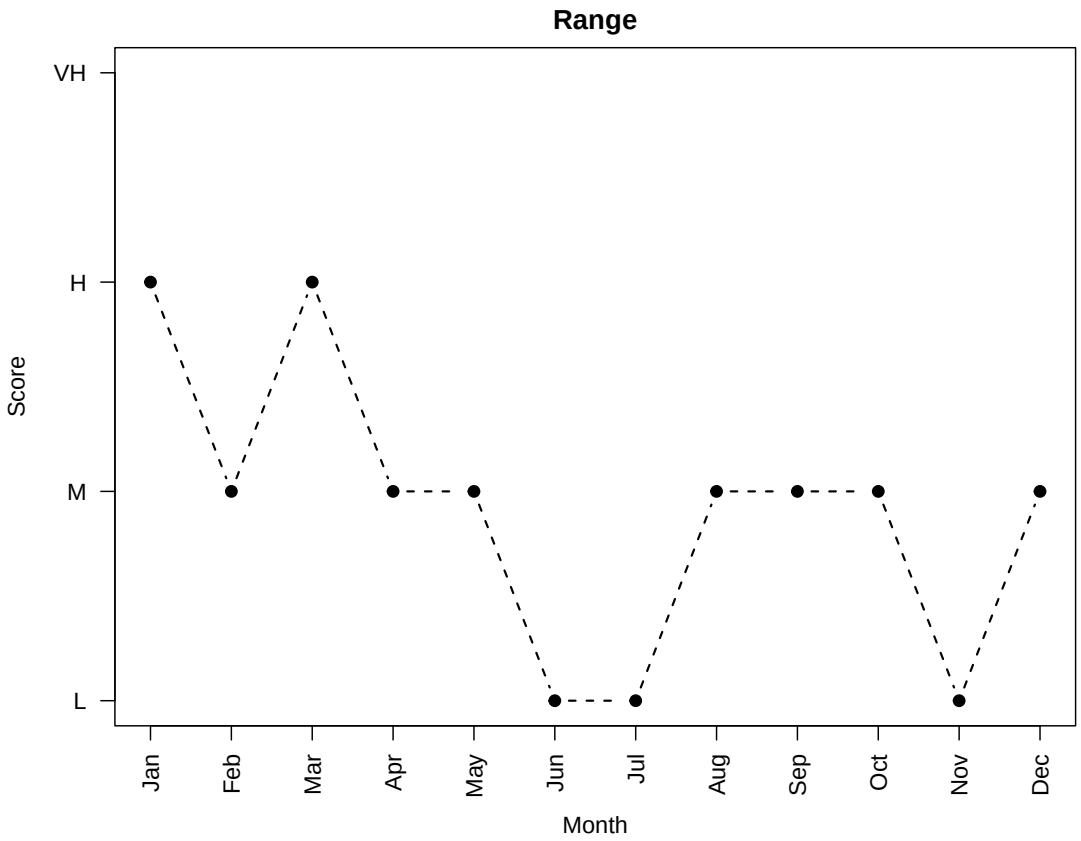


Figure 8: Timeseries of Total Exposure for Golden tilefish

Exposure due to Significant Factors

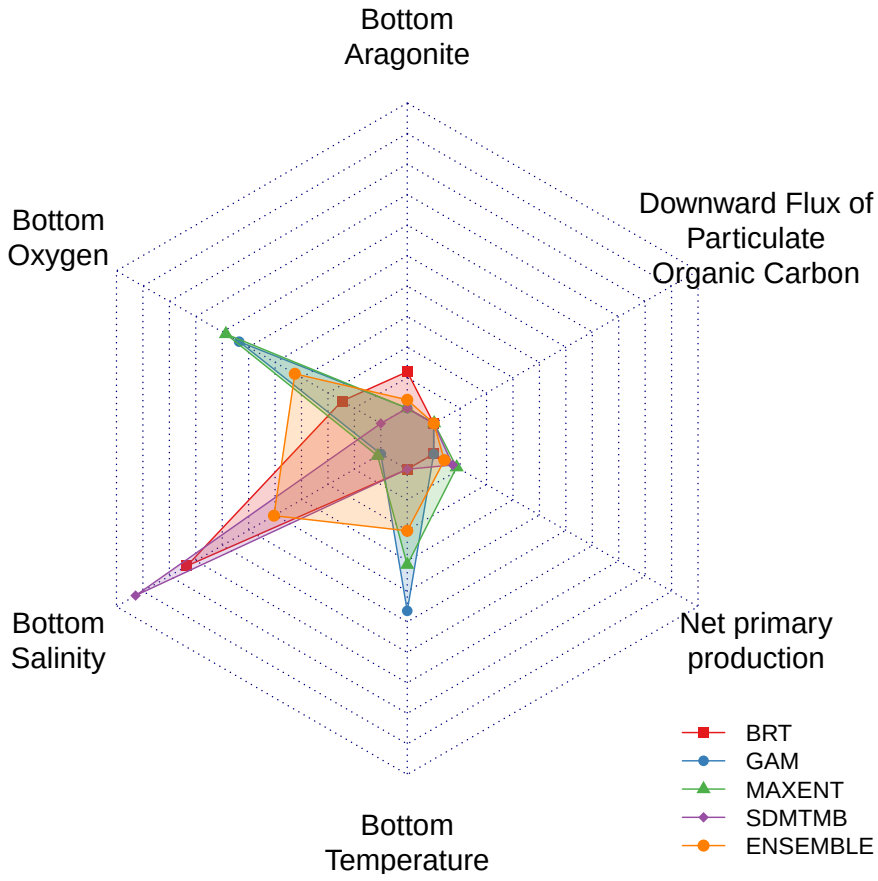


Figure 9: Dynamic Factors driving Golden tilefish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, oxygen, and salinity were the major dynamic drivers of golden tilefish distributions along the Northeast Shelf (Figure 9). Golden tilefish are very sensitive to temperature changes and water mass movements may play a large role in recruitment (Salois et al. 2024). When only these factors were used to quantify Exposure, the mean score decreased to Low (1.06 +/- 0.30) due to low bottom temperature exposure (Figure 10; Figure 11).

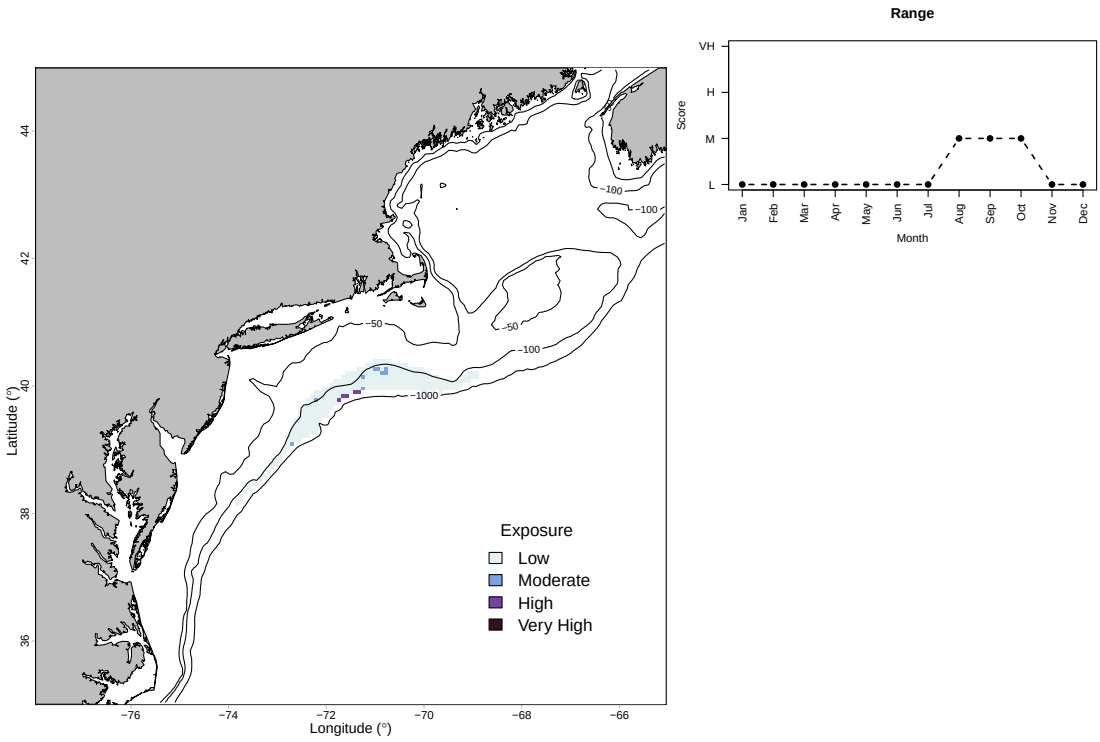
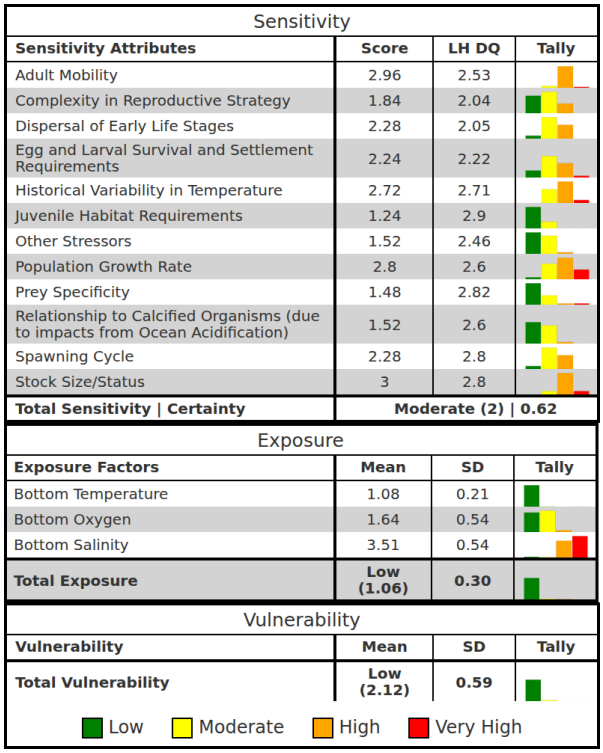


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Golden tilefish distributions



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	2.96	2.53	
Complexity in Reproductive Strategy	1.84	2.04	
Dispersal of Early Life Stages	2.28	2.05	
Egg and Larval Survival and Settlement Requirements	2.24	2.22	
Historical Variability in Temperature	2.72	2.71	
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Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.52	2.6	
Spawning Cycle	2.28	2.8	
Stock Size/Status	3	2.8	
Total Sensitivity | Certainty	**Moderate (2) | 0.62**		

Exposure

Exposure Factors	Mean	SD	Tally
Bottom Temperature	1.08	0.21	
Bottom Oxygen	1.64	0.54	
Bottom Salinity	3.51	0.54	
Total Exposure	**Low (1.06)**	**0.30**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (2.12)**	**0.59**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Golden tilefish, but with only significant Exposure factors and the resulting Vulnerability Scores.

References

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